

# Project Overview

This project is an **HR Analytics Dashboard** built using **Power BI** to analyze employee data and uncover insights related to workforce demographics, attrition, employee performance, and organizational trends. The dashboard provides an interactive and data-driven approach to help HR professionals make informed decisions and improve workforce management strategies.

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## Objectives

- Analyze employee demographics and workforce distribution.
  - Monitor employee attrition and retention trends.
  - Identify key factors influencing employee turnover.
  - Track department-wise and job-role-wise performance.
  - Provide actionable insights for HR decision-making.
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## Tools & Technologies Used

- **Power BI Desktop**
  - **Power Query** (Data Cleaning & Transformation)
  - **DAX (Data Analysis Expressions)**
  - **Data Modeling**
  - **Interactive Visualizations**
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## Dataset Information

The dataset contains employee-related information, including:

- Employee ID
- Age
- Gender
- Department
- Job Role
- Education Level
- Marital Status
- Salary
- Years at Company

- Job Satisfaction
  - Performance Rating
  - Attrition Status
  - Overtime Details
  - Other HR-related attributes
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## Dashboard Features

### Employee Overview

- Total Employees
- Active Employees
- Attrition Count
- Attrition Rate

### Demographic Analysis

- Employee distribution by Age Group
- Gender-wise distribution
- Education-wise analysis

### Department Analysis

- Department-wise employee count
- Job role distribution
- Department-wise attrition trends

### Attrition Analysis

- Attrition by Age Group
- Attrition by Salary Slab
- Attrition by Job Role
- Attrition by Overtime Status
- Years at Company vs Attrition

### Interactive Features

- Dynamic Filters/Slicers
  - Drill-down Analysis
  - Cross-filtering Visualizations
  - Interactive KPI Cards
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## Key Insights

- Identified departments and job roles with higher attrition rates.
  - Analyzed the relationship between employee age, experience, and turnover.
  - Determined the impact of overtime and salary on employee retention.
  - Highlighted demographic patterns influencing workforce distribution.
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## Business Recommendations

- Implement retention strategies for high-attrition departments.
  - Review compensation structures for employee segments with higher turnover.
  - Introduce employee engagement programs for at-risk groups.
  - Monitor overtime practices to reduce employee burnout.
  - Focus on career development and training initiatives.
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## Skills Demonstrated

- Data Cleaning and Transformation
- Data Modeling
- DAX Calculations
- Dashboard Design
- KPI Development
- Data Visualization
- Business Insight Generation
- HR Analytics and Reporting